

Project I: Historical cost of living index

By Merna Kamal, Guruvaraprasad Borra, Charlotte Lyda-Turner

Lorem Ipsum

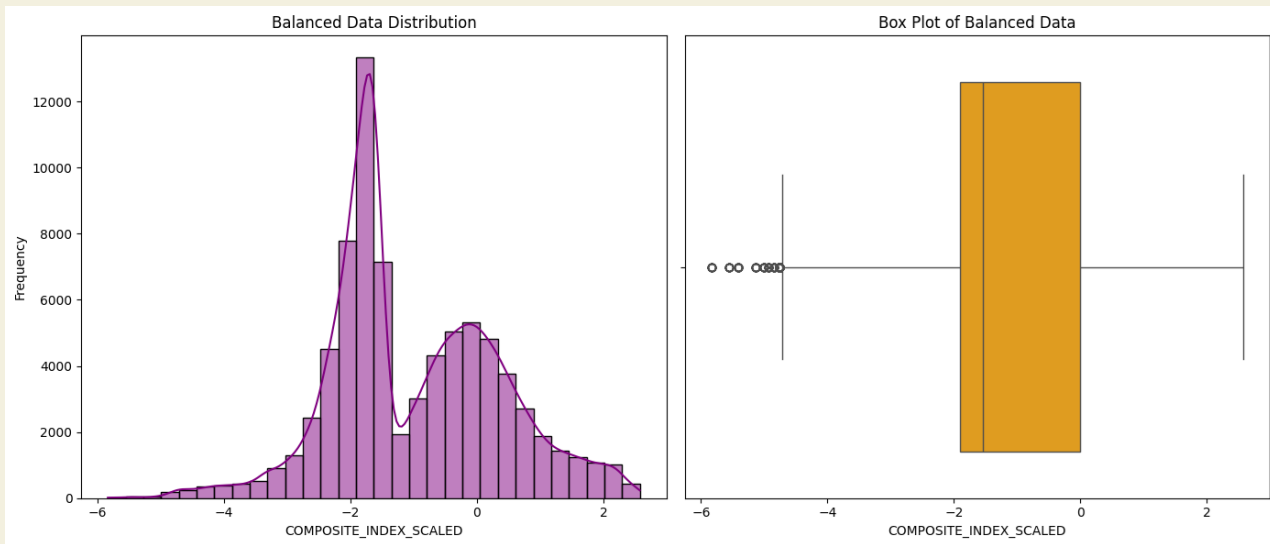


Project Overview

- The dataset for this project contained the Historical Cost of Living Index for various states from 1990 to 2024.
- **Goal:** The objective was to analyze the trends in the cost of living over time and identify the states that have experienced the most significant changes.
- **Machine Learning Approach:** Used linear regression to model the relationship between temporal features and the COLI.
- **Exploratory Data Analysis (EDA):** Employed to identify patterns, anomalies, and feature importance in the dataset.
- **Principal Component Analysis (PCA):** Applied to further analyze the dataset and reduce dimensionality.

Data Collection and Preprocessing

- Explored mean, median, and KNN imputation.
- Replaced missing values with the value of 5 neighbors.
- The dataset was imbalanced, with a skewness of 3.47. After using log transformation, Box_Cox, and oversampling the minority class techniques, skewness level reduced to 0.

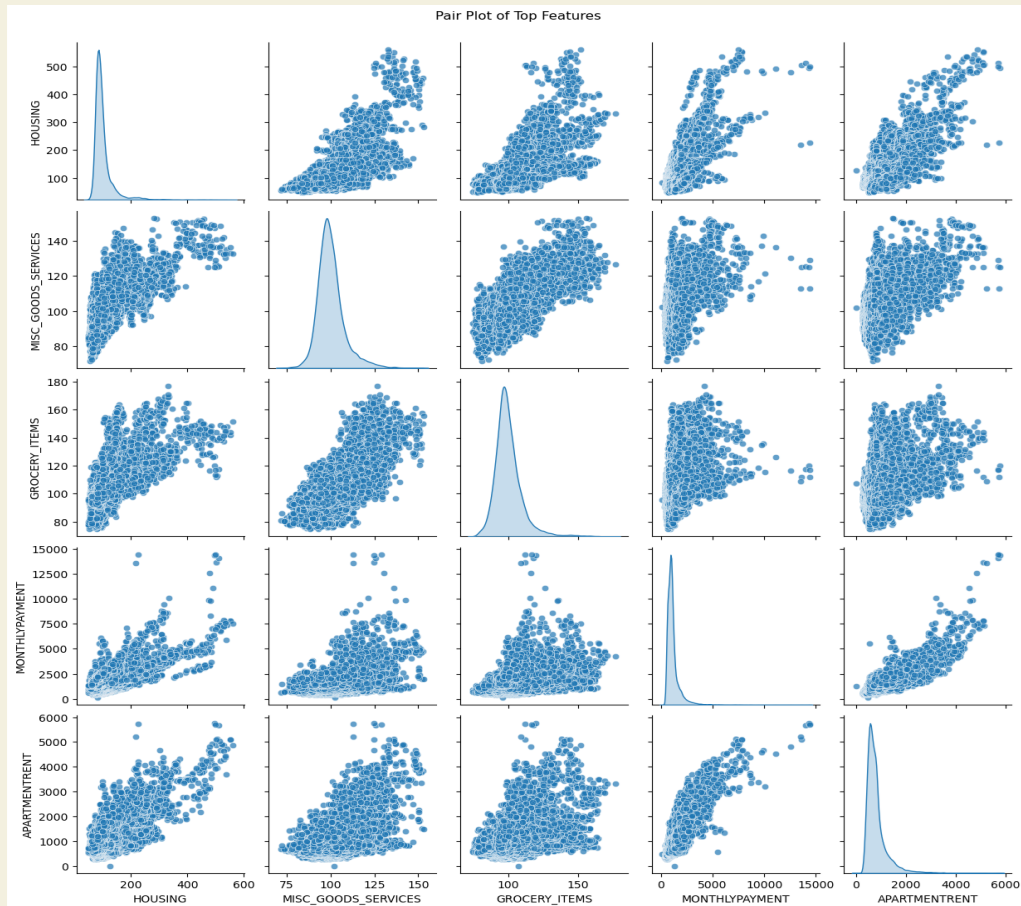


Feature Selection

Feature Extraction

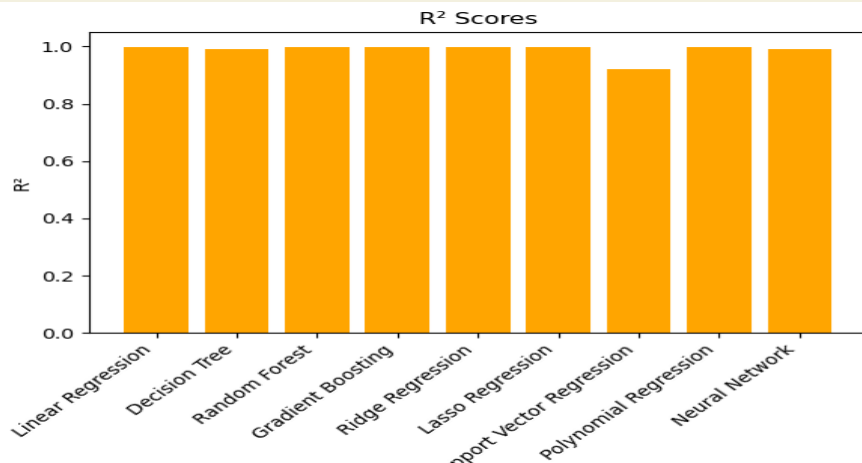
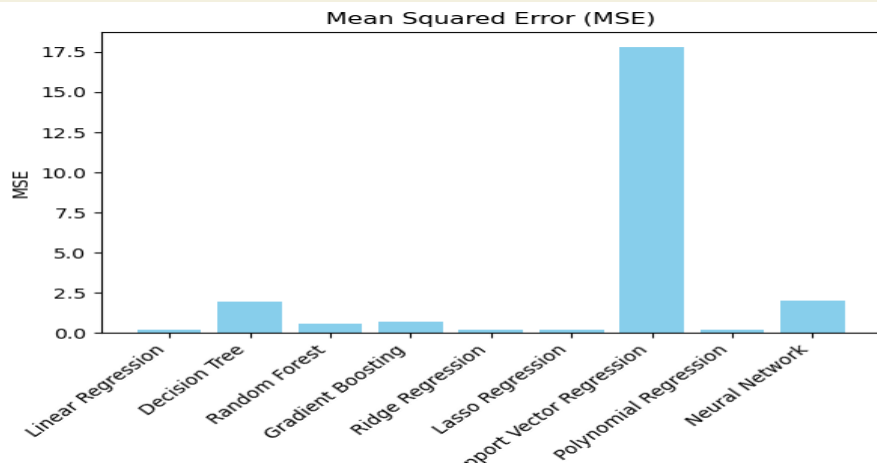
Top Correlated Features with COMPOSITE_INDEX:

COMPOSITE_INDEX		1.000000
HOUSING		0.964540
MISC_GOODS_SERVICES		0.792042
GROCERY_ITEMS		0.755311
MONTHLYPAYMENT		0.703036
MORTGAGERATE		0.003843
PRESCRIPTION_DRUG		0.003676
YEAR		0.000533
CITY_CODE		0.010354
STATE_CODE		-0.162798



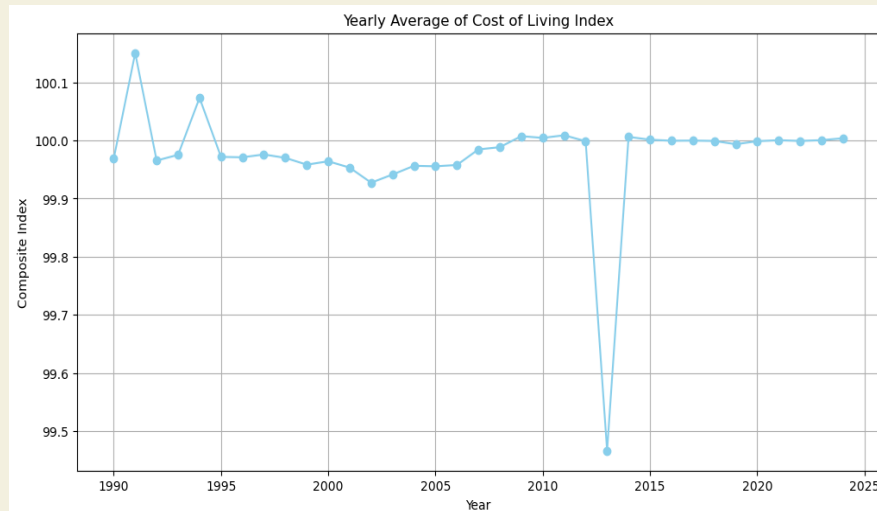
Model Selection and Justification

- Linear, Ridge, Lasso, Polynomial Regressions – simple, easily interpretable
- Decision Tree, Random Forest, Gradient Boosting – ability to handle non-linear relationships, robustness to overfitting
- Support Vector Regression – advantages in handling small, high-dimensional datasets
- Neural Network – capacity to model highly non-linear relationships



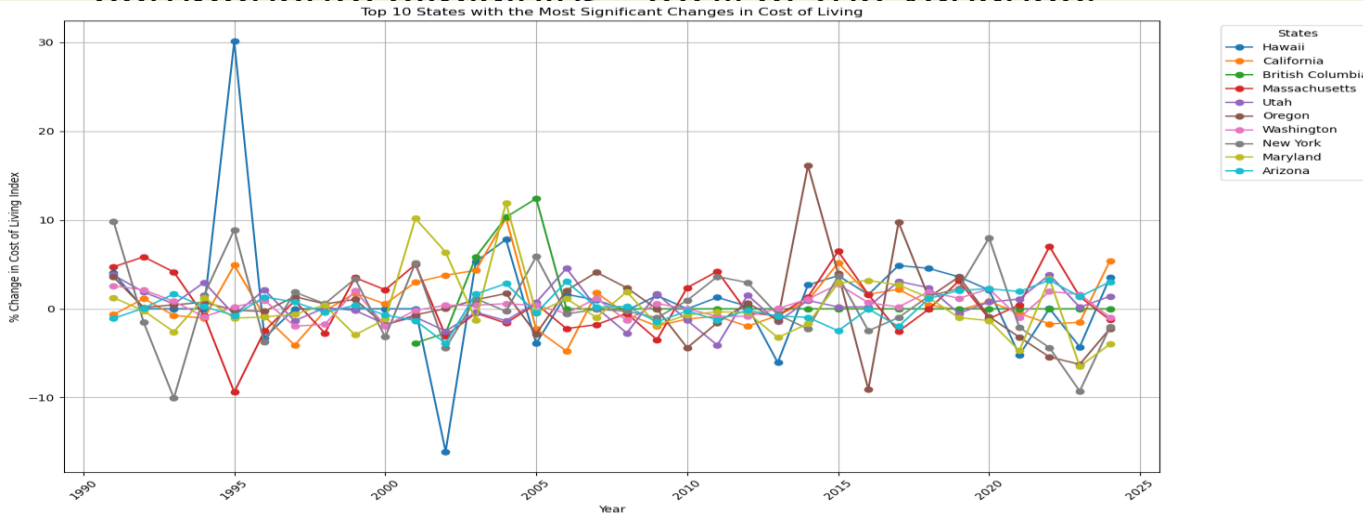
Final Results and Analysis

- Linear, Ridge, and Polynomial Regressions demonstrated outstanding predictive accuracy (R^2 scores above 0.9991)
 - Linear models well suited for problem
 - Near perfect predictive accuracy
- Random Forest and Gradient boosting also performed well (R^2 scores above 0.9969)
- Average cross-validation score across all models = 0.9989
 - consistency and robustness of results
- Models provided stable and reliable predictions



Discussion and Conclusion

- Most important features were HOUSING ($r \sim 0.88$), TRANSPORTATION ($r = 0.75$), UTILITIES ($r \sim 0.7$), YEAR ($r \sim 0.6-0.7$), HEALTH_CARE ($r \sim 0.6$), MISC_GOODS_SERVICES ($r \sim 0.65-0.7$), and GROCERY_ITEMS ($r \sim 0.5$).
- PCA reduced dimensionality efficiently, with the first two components explaining >99% of the variance.



Top 5 States with the Largest Positive Changes:

STATE_NAME

Hawaii 40.514767

California 22.055769

British Columbia 21.957061

Massachusetts 19.869188

Utah 14.819232

Top 5 States with the Largest Negative Changes:

STATE_NAME

Virgin Islands -25.409177

Michigan -14.232149

Connecticut -12.977698

Saskatoon -12.755343

West Virginia -12.483355

Cont. Discussion and conclusion

- Linear models (Linear Regression, Ridge Regression, Polynomial Regression) achieved R^2 values > 0.9991 , indicating strong linear relationships in the data.
- Non-linear models (Random Forest, Gradient Boosting) performed well but did not significantly outperform linear models ($R^2 > 0.9969$).
- Cross-validation scores averaged around 0.9989, confirming model robustness.

Future Work and Limitations

Real-World Applications:

- Findings can help policymakers and economists understand cost-of-living trends across states.
- Insights can guide decisions on cost-of-living adjustments, economic planning, and resource allocation.

Practical Implications:

- Accurate COLI trend predictions aid governments and businesses in preparing for economic shifts and responding to living cost fluctuations.

Limitations:

- The models used for this data is likely not as useful for places outside of the states surveyed.

Questions



THANK
YOU

Guruvaraprasad Borra
Charlotte Lyda-
Turner
Merna Bonoub